How Breast Cancer Survives Chemotherapy

Uncovering the "Emergency Response" Mechanism via RNA-seq

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The Mystery: Why Does Chemo Fail?

Chemotherapy is designed to kill.

Yet, many tumors survive the initial attack and develop resistance.

The Question:

"What is the very first thing the tumor does to survive?"

The Investigation: Rigorous Bioinformatics Pipeline

To ensure reproducibility, we utilized a standard **R-based workflow**:

1. **Data Processing:**
   - Quality Control & Alignment.
   - Filtering low-count genes.
2. **Statistical Analysis:**
   - Normalization: `DESeq2` (Median of Ratios).
   - Significance: Wald Test (FDR < 0.05).
3. **Visualization:**
   - Pathway Enrichment:

Pre $\rightarrow$

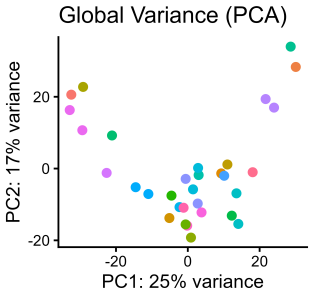
Post

The Shock: Treatment Rewires the Cell

Global Variance: The

treatment didn't just tweak the cell; it caused a massive **global shift**. (The

separation here proves the drug hit its target hard.)



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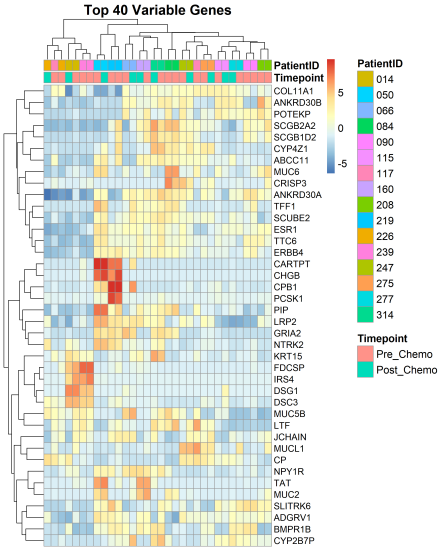
Post_Chemo:014
Post_Chemo:050
Post_Chemo:066
Post_Chemo:115
Post_Chemo:160
Post_Chemo:226
Post_Chemo:247
Post_Chemo:275
Post_Chemo:277
Post_Chemo:314
Pre_Chemo:014
Pre_Chemo:050
Pre_Chemo:066
Pre_Chemo:084
Pre_Chemo:090
Pre_Chemo:115
Pre_Chemo:117
Pre_Chemo:160
Pre_Chemo:208
Pre_Chemo:219
Pre_Chemo:226
Pre_Chemo:239
Pre_Chemo:247
Pre_Chemo:275
Pre
Pre

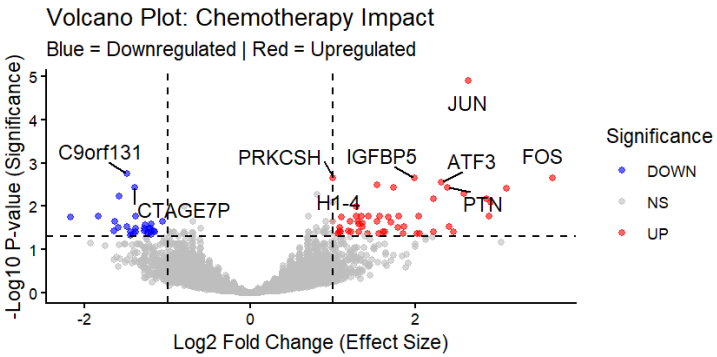
The Pattern: A Consistent Defense Plan

Top 40 Genes:

- This isn't random chaos.
- Every patient turned on the **same set of survival genes**.
- High consistency validates data quality.



The Culprit: Meet "AP-1"



Amidst the silence, two genes screamed out: **JUN & FOS** These

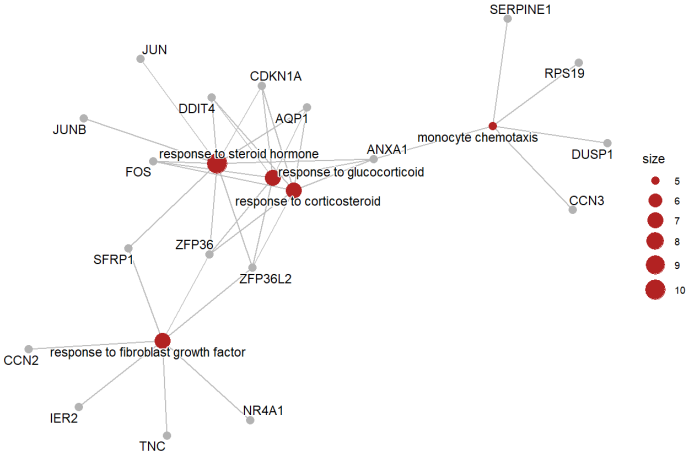
form the **AP-1 Complex**—the cell's "First Responder" to stress.

The Mechanism: The Emergency Shield

How does AP-1 work? It

acts as a central hub (Red Node), coordinating:

- Steroid Response
- Stress Signaling
- Survival



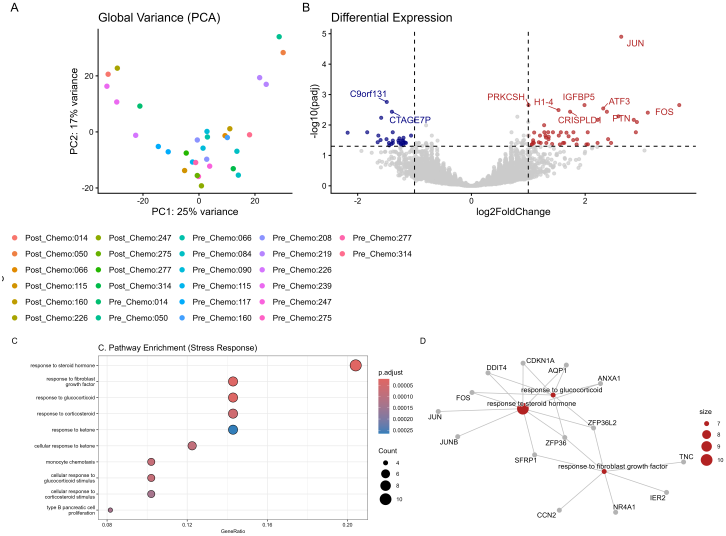
The Evidence: We Know This is Real

Our data was so sensitive, it detected two distinct signals:

1. **Hypoxia:** The tumor is starving for oxygen (The drug is working).
2. **Steroids:** We detected the *Dexamethasone* given to patients as pre-medication.

This confirms our "Stress Signal" is genuine.

The Big Picture: From Chaos to Mechanism



The Verdict: AP-1 is the Key

Conclusion

The tumor survives the initial chemotherapy shock by activating the **AP-1 Stress Complex (JUN/FOS)**.

The Opportunity:
If we can block AP-1 during the first dose, we might prevent the tumor from recovering.

Thank You

Questions?